

Storage in the Cloud TASK – 5

- 1. Create a new storage bucket in your preferred cloud provider (AWS S3, Google Cloud Storage, or Azure Blob Storage) and upload 3 sample files named song.mp3, food.jpg, and bill.pdf, simulating files from apps like Spotify, Zomato, and Paytm.**

Ans :

```
# Create a new S3 bucket
```

```
aws s3 mb s3://cloud-app-storage-demo --region ap-south-1
```

```
# Create sample files
```

```
echo "Sample Spotify song file" > song.mp3
```

```
echo "Sample Zomato food image file" > food.jpg
```

```
echo "Sample Paytm bill document" > bill.pdf
```

```
# Upload files to S3 bucket
```

```
aws s3 cp song.mp3 s3://cloud-app-storage-demo/
```

```
aws s3 cp food.jpg s3://cloud-app-storage-demo/
```

```
aws s3 cp bill.pdf s3://cloud-app-storage-demo/
```

```
# Verify uploaded files
```

```
aws s3 ls s3://cloud-app-storage-demo/
```

- 2. Configure a lifecycle policy on your storage bucket to automatically move food.jpg to a cheaper storage class (like S3 Glacier, Google Coldline, or Azure Cool) after 1 day and delete**

**bill.pdf after 3 days.

Hint:
Use your cloud provider's console or CLI to set up these rules,
and double-check the transition and expiration actions.**

Ans :

```
{
  "Rules": [
    {
      "ID": "Move-food-to-Glacier",
      "Status": "Enabled",
      "Filter": {
        "Prefix": "food.jpg"
      },
      "Transitions": [
        {
          "Days": 1,
          "StorageClass": "GLACIER"
        }
      ]
    },
    {
      "ID": "Delete-bill-after-3-days",
      "Status": "Enabled",
      "Filter": {
        "Prefix": "bill.pdf"
```

```
    },  
    "Expiration": {  
      "Days": 3  
    }  
  }  
]  
}
```

- 3. Monitor and document the status of your uploaded files over the next 24 hours: note which files have transitioned storage class or have been deleted according to your lifecycle policy.

Hint: Take a screenshot of the object details or use the CLI to confirm the changes.**

Ans :

```
{  
  "Rules": [  
    {  
      "ID": "Move-food-to-Glacier",  
      "Status": "Enabled",  
      "Filter": {  
        "Prefix": "food.jpg"  
      },  
      "Transitions": [  
        {  
          "Days": 1,  
          "StorageClass": "GLACIER"  
        }  
      ]  
    }  
  ]  
}
```

```

        "StorageClass": "GLACIER"
    }
]
},
{
    "ID": "Delete-bill-after-3-days",
    "Status": "Enabled",
    "Filter": {
        "Prefix": "bill.pdf"
    },
    "Expiration": {
        "Days": 3
    }
}
]
}

aws s3api put-bucket-lifecycle-configuration \
--bucket cloud-app-storage-demo \
--lifecycle-configuration file:///lifecycle.json

```

- 4. Compare the estimated monthly storage costs for your bucket with and without lifecycle policies enabled, using your cloud provider's pricing calculator. Write a short note on how lifecycle policies can help apps like Instagram or BookMyShow save money at scale.**

Ans :

Configuration	Storage Usage	Estimated Monthly Cost (INR)
Without Lifecycle Policy	All files stored in S3 Standard	₹_____
With Lifecycle Policy	Old files moved to Glacier / Unused files deleted	₹_____
Monthly Savings	—	₹_____

Short Note: Benefits of Lifecycle Policies

Lifecycle policies help applications automatically manage data storage by moving rarely accessed files to cheaper storage classes or deleting unnecessary data after a specific period.

For apps like Instagram:

- Millions of old photos and videos may not be accessed frequently.
- Older media can be moved to lower-cost storage tiers while keeping recent content in faster storage.
- This reduces storage expenses without losing data.

5. Use ChatGPT or another AI tool to generate a sample lifecycle policy JSON or YAML for a scenario where all .mp3 files are moved to cold storage after 7 days and deleted after 30 days. Paste the generated policy and explain any changes you would make for your own use case.

Ans :

{

```

"Rules": [
  {
    "ID": "MP3-Cold-Storage-and-Deletion",
    "Status": "Enabled",
    "Filter": {
      "Suffix": ".mp3"
    },
    "Transitions": [
      {
        "Days": 7,
        "StorageClass": "GLACIER"
      }
    ],
    "Expiration": {
      "Days": 30
    }
  }
]

```

- Explanation of Each Section
- Rules
 - Contains the lifecycle rules that define automated actions for objects in the S3 bucket.
- ID

- A unique identifier for the lifecycle rule.
 - Example: "MP3-Cold-Storage-and-Deletion".
- Status
 - Controls whether the rule is active.
 - "Enabled" means the lifecycle policy will run automatically.
- Filter
 - Defines which objects the rule applies to.
 - "Suffix": ".mp3" ensures the rule applies only to MP3 files.
- Transitions
 - Defines storage class changes.
 - "Days": 7 means MP3 files will move to Glacier after 7 days.
 - "StorageClass": "GLACIER" moves files to cheaper cold storage.
- Expiration
 - Defines when objects are permanently deleted.
 - "Days": 30 means MP3 files are deleted automatically after 30 days.